Final Project — Part 1

End-to-End Streaming Pipeline

This document presents the screenshot evidence for Part 1 of the GoIT Data Engineering Final Project. The pipeline reads athletes' bio data from MySQL, streams competition results from a Kafka topic, joins them, aggregates average height/weight by sport / medal / sex / country, then writes the aggregated records both to the output Kafka topic and to a MySQL table.

Stage 1: Read athlete_bio from MySQL (olympic_dataset), filter out rows with invalid height/weight.

Stage 2: Filter records where height or weight is empty or non-numeric.

Stage 3: Read athlete_event_results from MySQL, publish every row to the input Kafka topic oza_athlete_event_results.

Stage 3 (cont.): Read the input Kafka topic with Spark Structured Streaming, parse JSON to DataFrame columns.

Stage 4: Join the streaming events with the static bio DataFrame on athlete_id.

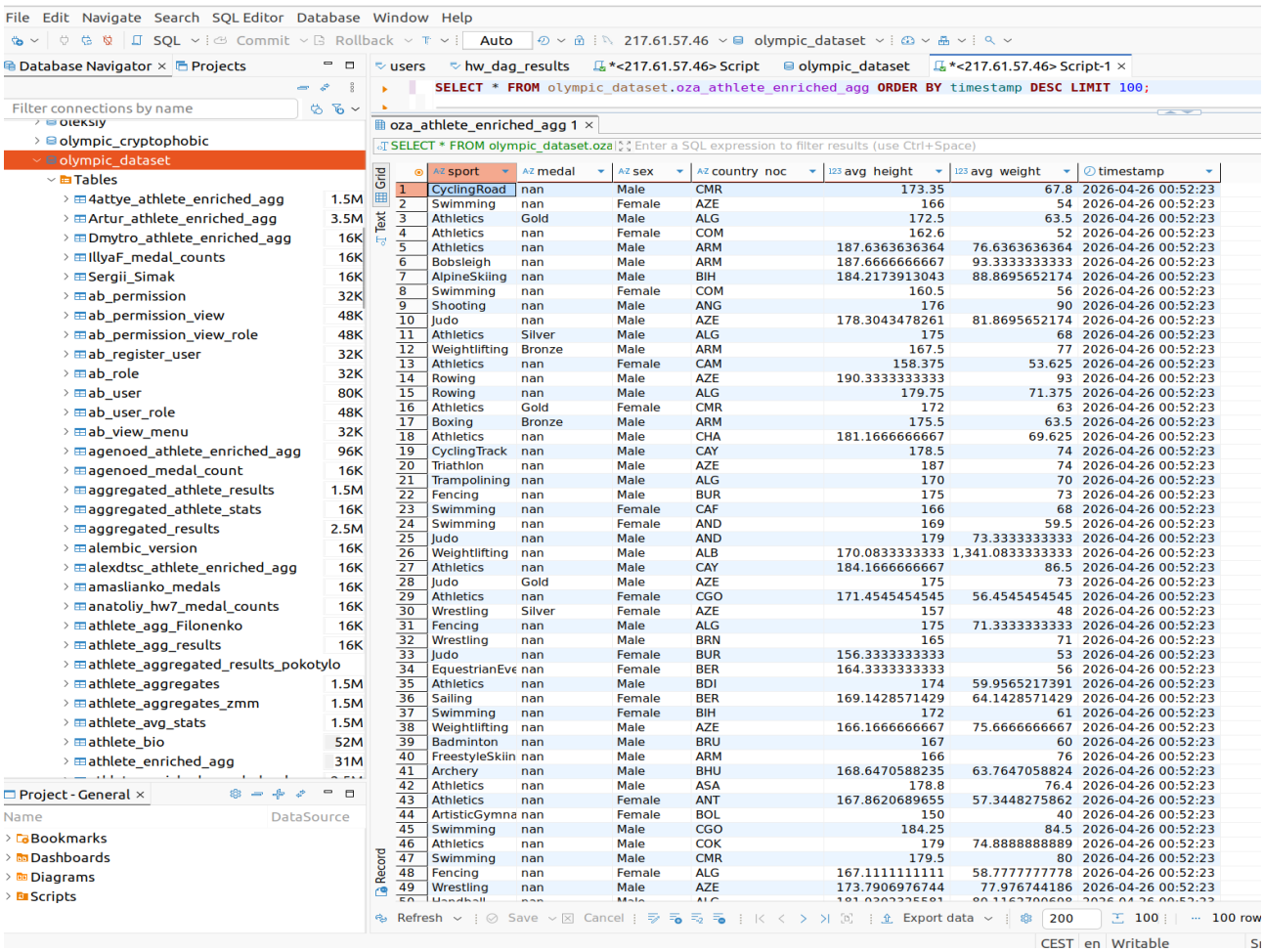
Stage 5: Compute avg_height and avg_weight grouped by sport, medal, sex, country_noc; add timestamp.

Stage 6a: Write each micro-batch (via foreachBatch) to the output Kafka topic oza_athlete_event_output.

Stage 6b: Write each micro-batch to the MySQL table oza_athlete_enriched_agg.

Screenshot 1 — MySQL Output Table (Stage 6b)

DBeaver query result for the MySQL table **olympic_dataset.oza_athlete_enriched_agg**. The table contains the final aggregated output written by the Spark Structured Streaming job after every micro-batch (Stage 6b). Columns: *sport, medal, sex, country_noc, avg_height, avg_weight, timestamp*. The timestamp column confirms when each aggregation was computed. 100 rows are shown; the table contained 473 rows at the time of the screenshot.



	sport	medal	sex	country_noc	avg_height	avg_weight	timestamp
1	CyclingRoad	nan	Male	CMR	173.35	67.8	2026-04-26 00:52:23
2	Swimming	nan	Female	AZE	166	54	2026-04-26 00:52:23
3	Athletics	Gold	Male	ALG	172.5	63.5	2026-04-26 00:52:23
4	Athletics	nan	Female	COM	162.6	52	2026-04-26 00:52:23
5	Athletics	nan	Male	ARM	187.6363636364	76.6363636364	2026-04-26 00:52:23
6	Bobsleigh	nan	Male	ARM	187.6666666667	93.3333333333	2026-04-26 00:52:23
7	AlpineSkiing	nan	Male	BIH	184.2173913043	88.8695652174	2026-04-26 00:52:23
8	Swimming	nan	Female	COM	160.5	56	2026-04-26 00:52:23
9	Shooting	nan	Male	ANG	176	90	2026-04-26 00:52:23
10	Judo	nan	Male	AZE	178.3043478261	81.8695652174	2026-04-26 00:52:23
11	Athletics	Silver	Male	ALG	175	68	2026-04-26 00:52:23
12	Weightlifting	Bronze	Male	ARM	167.5	77	2026-04-26 00:52:23
13	Athletics	nan	Female	CAM	158.375	53.625	2026-04-26 00:52:23
14	Rowing	nan	Male	AZE	190.3333333333	93	2026-04-26 00:52:23
15	Rowing	nan	Male	ALG	179.75	71.375	2026-04-26 00:52:23
16	Athletics	Gold	Female	CMR	172	63	2026-04-26 00:52:23
17	Boxing	Bronze	Male	ARM	175.5	63.5	2026-04-26 00:52:23
18	Athletics	nan	Male	CHA	181.1666666667	69.625	2026-04-26 00:52:23
19	CyclingTrack	nan	Male	CAY	178.5	74	2026-04-26 00:52:23
20	Triathlon	nan	Male	AZE	187	74	2026-04-26 00:52:23
21	Trampolining	nan	Male	ALG	170	70	2026-04-26 00:52:23
22	Fencing	nan	Male	BUR	175	73	2026-04-26 00:52:23
23	Swimming	nan	Female	CAF	166	68	2026-04-26 00:52:23
24	Swimming	nan	Female	AND	169	59.5	2026-04-26 00:52:23
25	Judo	nan	Male	AND	179	73.3333333333	2026-04-26 00:52:23
26	Weightlifting	nan	Male	ALB	170.0833333333	1,341.0833333333	2026-04-26 00:52:23
27	Athletics	nan	Male	CAY	184.1666666667	86.5	2026-04-26 00:52:23
28	Judo	Gold	Male	AZE	175	73	2026-04-26 00:52:23
29	Athletics	nan	Female	CGO	171.4545454545	56.4545454545	2026-04-26 00:52:23
30	Wrestling	Silver	Female	AZE	157	48	2026-04-26 00:52:23
31	Fencing	nan	Male	ALG	175	71.3333333333	2026-04-26 00:52:23
32	Wrestling	nan	Male	BRN	165	71	2026-04-26 00:52:23
33	Judo	nan	Female	BUR	156.3333333333	53	2026-04-26 00:52:23
34	EquestrianEve	nan	Female	BER	164.3333333333	56	2026-04-26 00:52:23
35	Athletics	nan	Male	BDI	174	59.9565217391	2026-04-26 00:52:23
36	Sailing	nan	Female	BER	169.1428571429	64.1428571429	2026-04-26 00:52:23
37	Swimming	nan	Female	BIH	172	61	2026-04-26 00:52:23
38	Weightlifting	nan	Male	AZE	166.1666666667	75.6666666667	2026-04-26 00:52:23
39	Badminton	nan	Male	BRU	167	60	2026-04-26 00:52:23
40	FreestyleSkiin	nan	Male	ARM	166	76	2026-04-26 00:52:23
41	Archery	nan	Male	BHU	168.6470588235	63.7647058824	2026-04-26 00:52:23
42	Athletics	nan	Male	ASA	178.8	76.4	2026-04-26 00:52:23
43	Athletics	nan	Female	ANT	167.8620689655	57.3448275862	2026-04-26 00:52:23
44	ArtisticGymna	nan	Female	BOL	150	40	2026-04-26 00:52:23
45	Swimming	nan	Male	CGO	184.25	84.5	2026-04-26 00:52:23
46	Athletics	nan	Male	COK	179	74.8888888889	2026-04-26 00:52:23
47	Swimming	nan	Male	CMR	179.5	80	2026-04-26 00:52:23
48	Fencing	nan	Female	ALG	167.1111111111	58.7777777778	2026-04-26 00:52:23
49	Wrestling	nan	Male	AZE	173.7906976744	77.976744186	2026-04-26 00:52:23

Fig. 1 — SELECT * FROM olympic_dataset.oza_athlete_enriched_agg ORDER BY timestamp DESC LIMIT 100;

Screenshot 2 — Kafka Output Topic Consumer (Stage 6a)

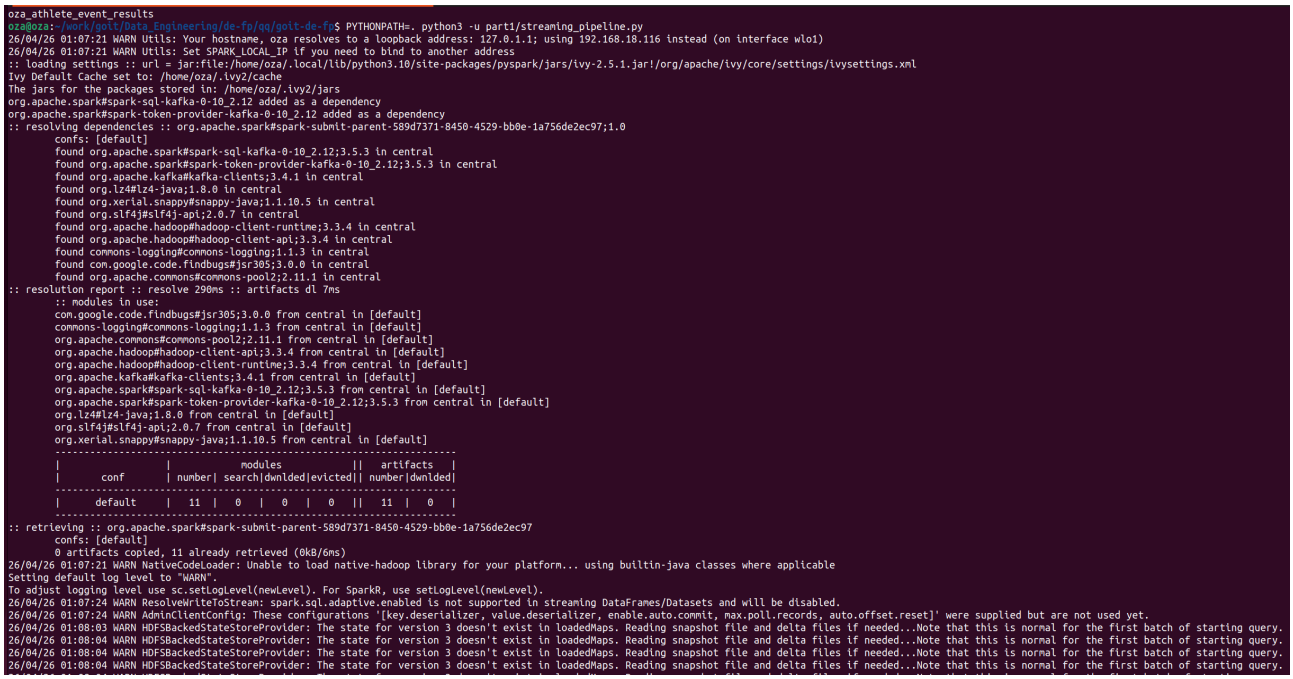
Terminal output of **part1/output_consumer.py** subscribed to the Kafka topic **oza_athlete_event_output**. Each line shows a message key (UUID) and a JSON value containing the aggregated record: *sport, medal, sex, country_noc, avg_height, avg_weight, timestamp*. This confirms that Stage 6a successfully streamed aggregated data to the Kafka output topic.

```
oza@oza:~/work/goit/Data_Engineering/de-fp/oq/goit-de-fp$ PYTHONPATH=. python3 -u part1/output_consumer.py
Subscribed to topic: oza_athlete_event_output
key=26cc5c69-8bbb-4bb1-9994-ac07eb513d82 value={'sport': 'Swimming', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'ANZ', 'avg_height': 170.0, 'avg_weight': 65.0, 'timestamp': '2026-04-26T01:07:54.796+02:00'}
key=26cc5c69-8bbb-4bb1-9994-ac07eb513d82 value={'sport': 'Athletics', 'medal': 'Bronze', 'sex': 'Male', 'country_noc': 'ANZ', 'avg_height': 184.0, 'avg_weight': 76.0, 'timestamp': '2026-04-26T01:07:54.796+02:00'}
key=26cc5c69-8bbb-4bb1-9994-ac07eb513d82 value={'sport': 'Boxing', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'ANZ', 'avg_height': 167.0, 'avg_weight': 62.0, 'timestamp': '2026-04-26T01:07:54.796+02:00'}
key=26cc5c69-8bbb-4bb1-9994-ac07eb513d82 value={'sport': 'Rugby', 'medal': 'Gold', 'sex': 'Male', 'country_noc': 'ANZ', 'avg_height': 175.0, 'avg_weight': 75.0, 'timestamp': '2026-04-26T01:07:54.796+02:00'}
key=26cc5c69-8bbb-4bb1-9994-ac07eb513d82 value={'sport': 'Swimming', 'medal': 'Silver', 'sex': 'Male', 'country_noc': 'ANZ', 'avg_height': 170.0, 'avg_weight': 65.0, 'timestamp': '2026-04-26T01:07:54.796+02:00'}
key=26cc5c69-8bbb-4bb1-9994-ac07eb513d82 value={'sport': 'Swimming', 'medal': 'Bronze', 'sex': 'Male', 'country_noc': 'ANZ', 'avg_height': 170.0, 'avg_weight': 65.0, 'timestamp': '2026-04-26T01:07:54.796+02:00'}
key=26cc5c69-8bbb-4bb1-9994-ac07eb513d82 value={'sport': 'Athletics', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'ANZ', 'avg_height': 170.83333333333334, 'avg_weight': 66.0, 'timestamp': '2026-04-26T01:07:54.796+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'FreestyleSkiing', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'ARM', 'avg_height': 166.0, 'avg_weight': 76.0, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Athletics', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'BDI', 'avg_height': 174.0, 'avg_weight': 59.95652173913044, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Wrestling', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'AZE', 'avg_height': 173.7906976744186, 'avg_weight': 77.97674418604652, 'timestamp': '2026-04-26T01:08:12.
584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'CanoeSprint', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'BOL', 'avg_height': 184.0, 'avg_weight': 85.0, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Shooting', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'BOL', 'avg_height': 173.888888888888889, 'avg_weight': 77.16666666666667, 'timestamp': '2026-04-26T01:08:12.
584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Fencing', 'medal': 'nan', 'sex': 'Female', 'country_noc': 'ALG', 'avg_height': 167.11111111111111, 'avg_weight': 58.77777777777778, 'timestamp': '2026-04-26T01:08:12
.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Athletics', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'ASA', 'avg_height': 178.8, 'avg_weight': 76.4, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'CyclingRoad', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'BAR', 'avg_height': 173.71428571428572, 'avg_weight': 69.71428571428571, 'timestamp': '2026-04-26T01:08:
12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Athletics', 'medal': 'nan', 'sex': 'Female', 'country_noc': 'CHA', 'avg_height': 165.55555555555554, 'avg_weight': 57.77777777777778, 'timestamp': '2026-04-26T01:08:
12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Swimming', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'ASA', 'avg_height': 181.5, 'avg_weight': 80.5, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Athletics', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'AND', 'avg_height': 174.55555555555554, 'avg_weight': 63.666666666666664, 'timestamp': '2026-04-26T01:08:1
2.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Sailing', 'medal': 'nan', 'sex': 'Female', 'country_noc': 'BER', 'avg_height': 169.14285714285714, 'avg_weight': 64.14285714285714, 'timestamp': '2026-04-26T01:08:12
.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Swimming', 'medal': 'nan', 'sex': 'Female', 'country_noc': 'CMR', 'avg_height': 170.0, 'avg_weight': 55.0, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Swimming', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'ALB', 'avg_height': 188.5, 'avg_weight': 83.25, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Taekwondo', 'medal': 'nan', 'sex': 'Female', 'country_noc': 'CPV', 'avg_height': 169.0, 'avg_weight': 49.0, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Swimming', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'CGO', 'avg_height': 184.25, 'avg_weight': 84.5, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Athletics', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'ALG', 'avg_height': 175.88695652173914, 'avg_weight': 66.70434782608696, 'timestamp': '2026-04-26T01:08:12
.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'EquestrianJumping', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'AZE', 'avg_height': 178.0, 'avg_weight': 74.0, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Weightlifting', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'AZE', 'avg_height': 166.16666666666666, 'avg_weight': 75.66666666666667, 'timestamp': '2026-04-26T01:0
8:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Judo', 'medal': 'nan', 'sex': 'Female', 'country_noc': 'CHA', 'avg_height': 175.0, 'avg_weight': 70.0, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Swimming', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'AZE', 'avg_height': 181.2, 'avg_weight': 73.4, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Archery', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'BHU', 'avg_height': 168.64705882352942, 'avg_weight': 63.764705882352294, 'timestamp': '2026-04-26T01:08:12.5
84+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Athletics', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'COK', 'avg_height': 179.0, 'avg_weight': 74.88888888888889, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Swimming', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'CAM', 'avg_height': 172.4375, 'avg_weight': 68.125, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Athletics', 'medal': 'nan', 'sex': 'Female', 'country_noc': 'ANT', 'avg_height': 167.86206896551724, 'avg_weight': 57.3448275862069, 'timestamp': '2026-04-26T01:08:1
2.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Athletics', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'CGO', 'avg_height': 176.67391304347825, 'avg_weight': 69.80434782608695, 'timestamp': '2026-04-26T01:08:12
.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'CyclingRoad', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'ARU', 'avg_height': 172.66666666666666, 'avg_weight': 73.66666666666667, 'timestamp': '2026-04-26T01:08:
12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Wrestling', 'medal': 'Bronze', 'sex': 'Male', 'country_noc': 'AZE', 'avg_height': 176.375, 'avg_weight': 87.5, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Swimming', 'medal': 'nan', 'sex': 'Female', 'country_noc': 'BIH', 'avg_height': 172.0, 'avg_weight': 61.0, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'CanoeSlalom', 'medal': 'nan', 'sex': 'Female', 'country_noc': 'COK', 'avg_height': 160.0, 'avg_weight': 62.0, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Boxing', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'COK', 'avg_height': 169.33333333333334, 'avg_weight': 56.0, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Badminton', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'ALG', 'avg_height': 182.0, 'avg_weight': 82.0, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Weightlifting', 'medal': 'nan', 'sex': 'Female', 'country_noc': 'COK', 'avg_height': 165.0, 'avg_weight': 101.0, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Biathlon', 'medal': 'nan', 'sex': 'Female', 'country_noc': 'AND', 'avg_height': 160.0, 'avg_weight': 49.0, 'timestamp': '2026-04-26T01:08:12.584+02:00'}
key=f210267d-9e06-4b0f-b2be-6b7c4b49baba value={'sport': 'Athletics', 'medal': 'nan', 'sex': 'Male', 'country_noc': 'BAN', 'avg_height': 169.7826086956522, 'avg_weight': 63.391304347826086, 'timestamp': '2026-04-26T01:08:12
.584+02:00'}
```

Fig. 2 — Messages received from topic oza_athlete_event_output via output_consumer.py.

Screenshot 3 — Spark Structured Streaming Pipeline Running

Terminal output of **part1/streaming_pipeline.py**. The log shows Spark successfully resolving the Kafka connector packages (*spark-sql-kafka-0-10_2.12:3.5.3*), starting the SparkSession in local mode, and entering the streaming loop (awaitTermination). The WARN lines about HDFSBackedStateStore are informational — they indicate the first batch checkpoint creation and are expected on a fresh start.



```
oza_athlete_event_results
oza@oza:~/serial/projects/Engineering/de-fp/qq/qoit-de-fp$ PYTHONPATH=. python3 -u part1/streaming_pipeline.py
26/04/26 01:07:21 WARN Utils: Your hostname, oza resolves to a loopback address: 127.0.1.1; using 192.168.18.116 instead (on interface wlo1)
26/04/26 01:07:21 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
:: loading settings :: url = jar:file:/home/oza/.local/lib/python3.10/site-packages/pyspark/jars/ivy-2.5.1.jar!/org/apache/ivy/core/settings/ivysettings.xml
Ivy Default Cache set to: /home/oza/.ivy2/cache
The jars for the packages stored in: /home/oza/.ivy2/jars
org.apache.spark#spark-sql-kafka-0-10_2.12 added as a dependency
org.apache.spark#spark-token-provider-kafka-0-10_2.12 added as a dependency
:: resolving dependencies :: org.apache.spark#spark-submit-parent-589d7371-8450-4529-bb0e-1a756de2ec97;1.0
	confs: [default]
	found org.apache.spark#spark-sql-kafka-0-10_2.12;3.5.3 in central
	found org.apache.spark#spark-token-provider-kafka-0-10_2.12;3.5.3 in central
	found org.apache.kafka#kafka-clients;3.4.1 in central
	found org.lz4#lz4-java;1.8.0 in central
	found org.xerial.snappy#snappy-java;1.1.10.5 in central
	found org.slf4j#slf4j-api;2.0.7 in central
	found org.apache.hadoop#hadoop-client-runtime;3.3.4 in central
	found org.apache.hadoop#hadoop-client-api;3.3.4 in central
	found commons-logging#commons-logging;1.1.3 in central
	found com.google.code.findbugs#jsr305;3.0.0 in central
	found org.apache.commons#commons-pool2;2.11.1 in central
:: resolution report :: resolve 290ms :: artifacts dl 7ms
	:: modules in use:
	com.google.code.findbugs#jsr305;3.0.0 from central in [default]
	commons-logging#commons-logging;1.1.3 from central in [default]
	org.apache.commons#commons-pool2;2.11.1 from central in [default]
	org.apache.hadoop#hadoop-client-api;3.3.4 from central in [default]
	org.apache.hadoop#hadoop-client-runtime;3.3.4 from central in [default]
	org.apache.kafka#kafka-clients;3.4.1 from central in [default]
	org.apache.spark#spark-sql-kafka-0-10_2.12;3.5.3 from central in [default]
	org.apache.spark#spark-token-provider-kafka-0-10_2.12;3.5.3 from central in [default]
	org.lz4#lz4-java;1.8.0 from central in [default]
	org.slf4j#slf4j-api;2.0.7 from central in [default]
	org.xerial.snappy#snappy-java;1.1.10.5 from central in [default]
	---------------------------------------------------------------------
	|                  |            modules            ||   artifacts   |
	|       conf       | number| search|dwnlded|evicted|| number|dwnlded|
	---------------------------------------------------------------------
	|      default     |   11  |   0   |   0   |   0   ||   11  |   0   |
	---------------------------------------------------------------------
:: retrieving :: org.apache.spark#spark-submit-parent-589d7371-8450-4529-bb0e-1a756de2ec97
	confs: [default]
	0 artifacts copied, 11 already retrieved (0kB/6ms)
26/04/26 01:07:21 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/04/26 01:07:24 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
26/04/26 01:07:24 WARN AdminClientConfig: These configurations '[key.deserializer, value.deserializer, enable.auto.commit, max.poll.records, auto.offset.reset]' were supplied but are not used yet.
26/04/26 01:08:03 WARN HDFSBackedStateStoreProvider: The state for version 3 doesn't exist in loadedMaps. Reading snapshot file and delta files if needed...Note that this is normal for the first batch of starting query.
26/04/26 01:08:04 WARN HDFSBackedStateStoreProvider: The state for version 3 doesn't exist in loadedMaps. Reading snapshot file and delta files if needed...Note that this is normal for the first batch of starting query.
26/04/26 01:08:04 WARN HDFSBackedStateStoreProvider: The state for version 3 doesn't exist in loadedMaps. Reading snapshot file and delta files if needed...Note that this is normal for the first batch of starting query.
26/04/26 01:08:04 WARN HDFSBackedStateStoreProvider: The state for version 3 doesn't exist in loadedMaps. Reading snapshot file and delta files if needed...Note that this is normal for the first batch of starting query.
```

Fig. 3 — streaming_pipeline.py startup: Spark resolves Kafka packages and enters streaming mode.